

Modelling Data

Task 2

Introduction

Overview

This project investigates a large-scale automotive insurance dataset comprising 268,255 records related to vehicle repair claims. The data captures the comprehensive economic lifecycle of an insurance claim, spanning from initial vehicle specifications such as manufacturer, age and mileage to technical repair details, including labor hours, complexity and final costs. A critical feature of this dataset is the inclusion of an `anomaly_present` flag, which identifies claims deviating from standard patterns likely representing financial leakage or fraudulent activity.

The objective of this assessment is to transition from initial exploratory data analysis (EDA) to advanced predictive and descriptive modelling.

The primary aim of this project is to develop a data driven framework that identifies high risk segments and predicts financial anomalies within the automotive insurance sector.

Reflection on task 1 feedback and rectification

Following the task 1 submission, several areas for improvement were identified. This section details the critical reflection on the points and step to be taken to rectify them for task 2.

Accuracy of data interpretation

A significant contradiction was noted in the task 1 analysis regarding the electric vehicle (EV) anomaly rates. The initial analysis suggested a stable anomaly rate despite the visual evidence showing nearly 100% anomalies for electric vehicles. This analysis declaration had to be revisited because it did not truly represent the data and could leave room for exploration.

We will frame this as a lesson in data fidelity as the initial analysis failed to align the narrative with the visual evidence and the rectification is a commitment to a grounded, evidence-based interpretation.

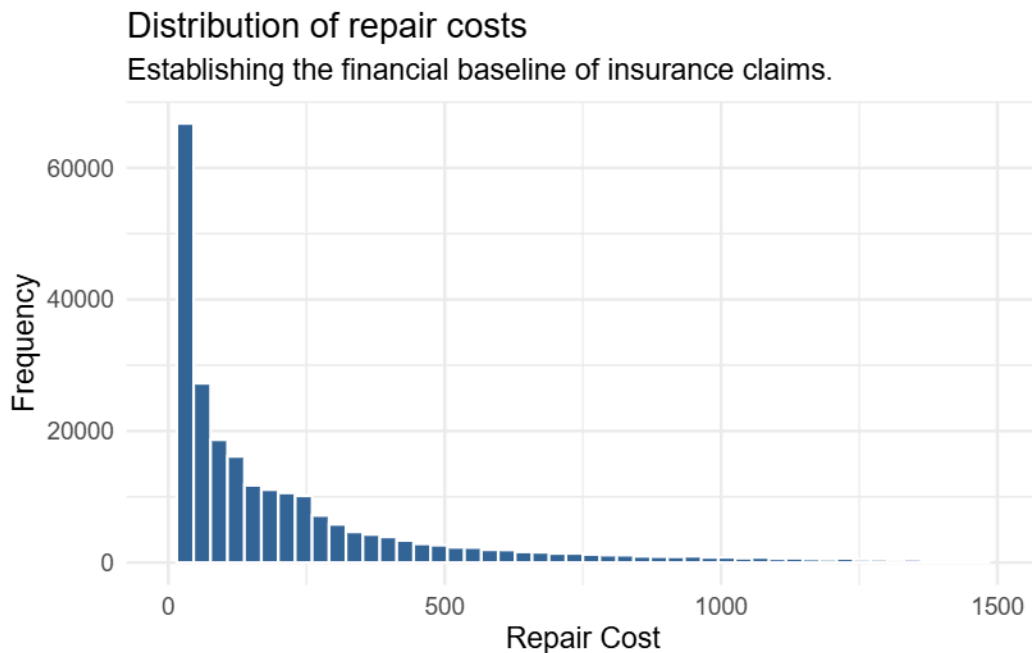
Comprehensive exploratory data analysis

Explanation and exploration of single variables was lacking in the EDA which created a gap in understanding the variables present to test plausible relationships with target variables. I have expanded the EDA to include the thorough analysis of major primary features which will aid the dataset's characteristics for research question development.

We will acknowledge that jumping straight into relationships without explaining the variables individually created a contextual vacuum. The fix is then to establish univariate baselines before bivariate relationships.

Supplemental EDA

Distribution of repair costs

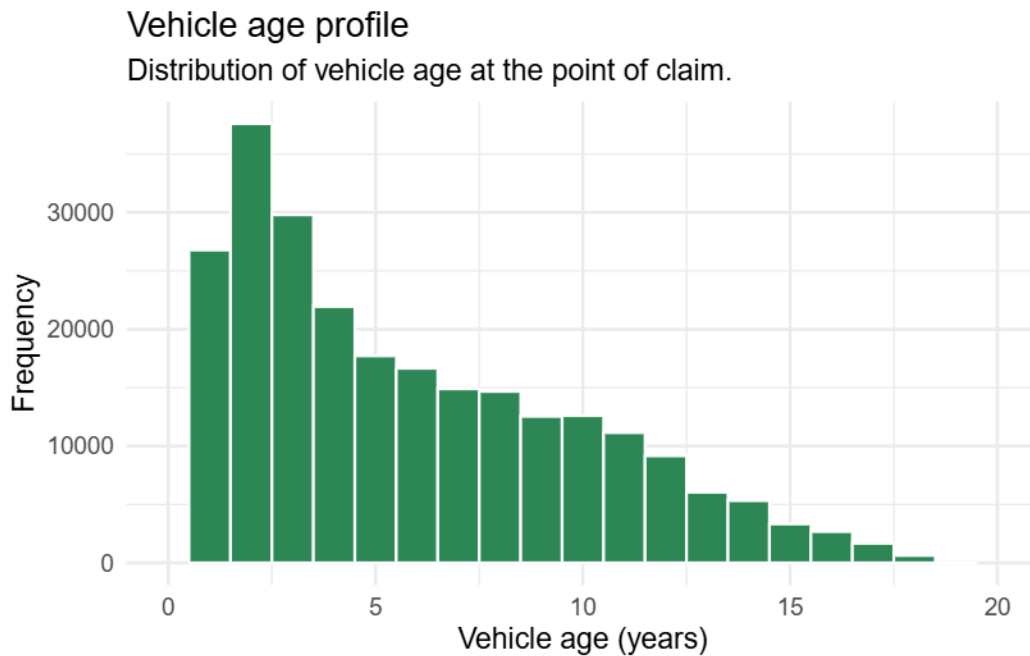


The graph demonstrates a positive skew where the overwhelming majority of repair claims are clustered at the extreme low end of the price spectrum. This indicates that a substantial proportion of insurance claims involve minor repairs likely those costing less than £1,000. While claims exceeding £5,000 or £10,000 exist, they are statistically rare compared to the

high volume of standard repairs meaning that the economic impact of claims varies from very small amounts (minimal structural damage) to costs that could potentially exceed the total market value of the vehicle.

The high-cost outliers represent the areas of greatest financial risk for an insurer where the potential for financial leakage such as over-estimated repairs or inflated parts costs is highly present. Understanding the distribution is a prerequisite for determining whether specific high costing repairs are a legitimate consequence of vehicle damage or a suspicious deviation from the norm.

Vehicle age profile



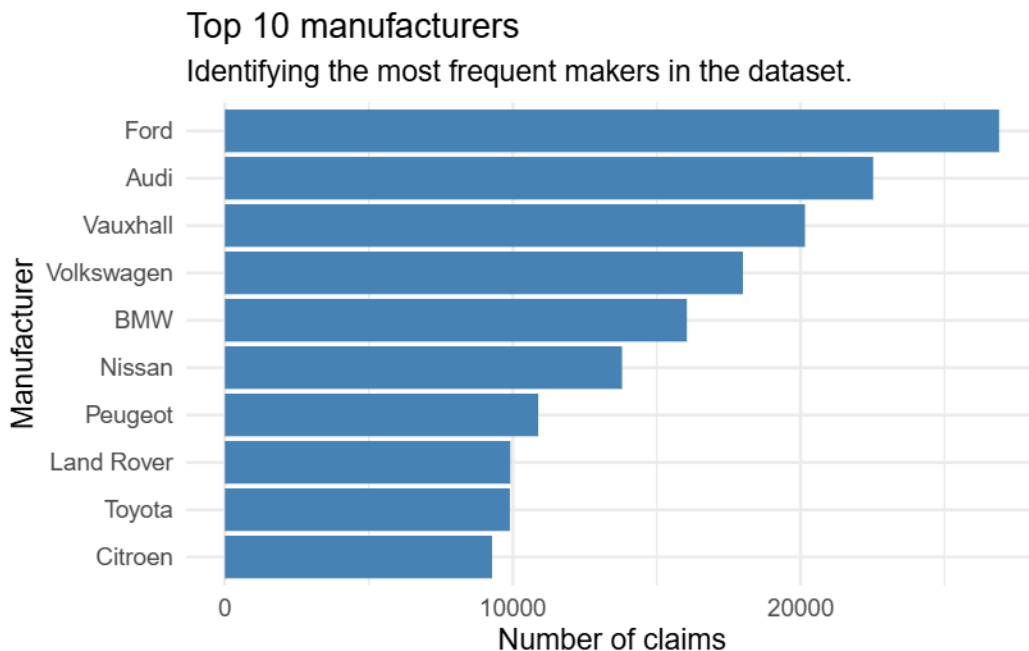
The distribution exhibits a spike for vehicles aged between 0 and 2 years indicating that the largest segment of the claims dataset consists of brand-new or late-model vehicles. There is a decline in the number of claims as vehicles age beyond 5 years suggesting that the majority of claims in this portfolio are from relatively modern vehicles.

Vehicle age is a primary determinant of mechanical integrity and repair logic and for the newer cars, repairs often involve expensive modern sensors and manufacturer original parts which naturally drives up the baseline claims cost. For the older vehicles, even minor damage may be flagged as a total loss because the repair cost quickly exceeds the vehicle's depreciated market value. Understanding the age distribution is crucial for identifying if anomalies are driven by high tech repair costs in new cars or uneconomic write-off thresholds in older ones

Manufacturer frequency (maker)

```
Plot3 <- ds5_final %>%  
  count(maker) %>%  
  slice_max(n, n = 10) %>%  
  ggplot(aes(x = reorder(maker, n), y = n)) +  
  geom_col(fill = "steelblue") +  
  coord_flip() +  
  theme_minimal() +  
  labs(title = "Top 10 manufacturers",  
        subtitle = "Identifying the most frequent makers in the dataset.",  
        x = "Manufacturer",  
        y = "Number of claims")
```

Plot3



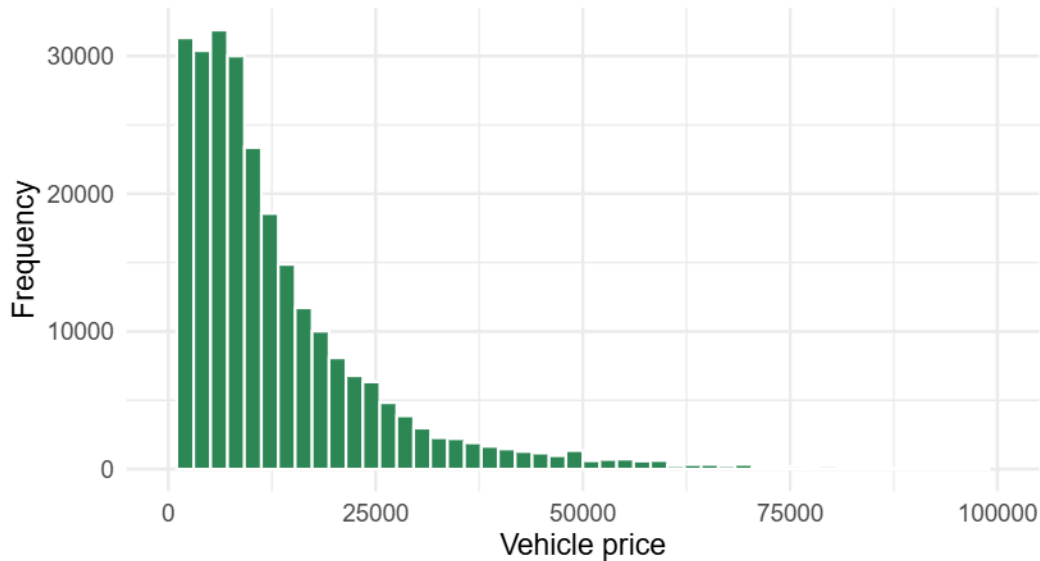
The distribution includes a mix of economy brands (Ford, Vauxhall, Peugeot) and premium manufacturers (BMW, Audi, Mercedes-Benz). Ford is the most frequent manufacturer in the dataset by a considerable margin followed by Audi, Vauxhall and Volkswagen.

Ford having the highest volume of claims, we would naturally expect it to have the highest number of anomaly claims. If a lower-volume brand like BMW shows a higher percentage of anomaly claims than Ford, it suggests a brand specific risk rather than a simple volume-based trend. This manufacturer profile ensures that the later findings are not biased by the sheer size of specific sub-groups within the data.

Costs of vehicle (price)

Distribution of vehicle prices

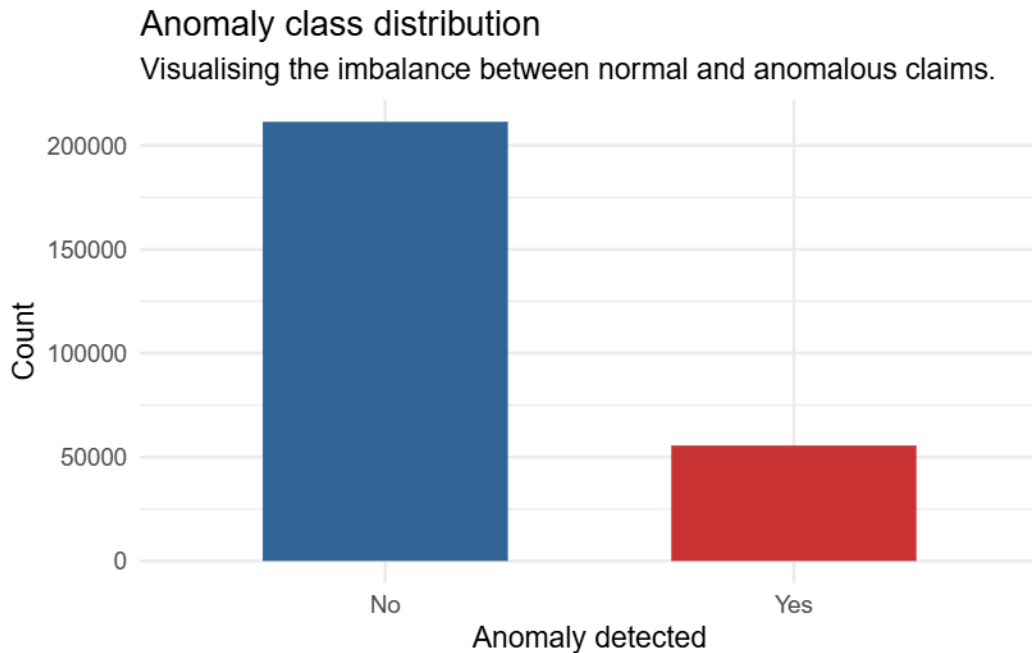
Visualising the market value of vehicles in the claims dataset.



The vast majority of vehicles are concentrated at the lower end of the valuation scale with the highest frequency of vehicles is found in the £5,000 to £15,000 range. This aligns with the top 10 manufacturers data suggesting a portfolio dominated by standard affordable consumer models from brands like Ford and Vauxhall. There are visible data points at the far right of the graph exceeding £100,000 which represent luxury and high end supercar assets or likely data entry errors that require careful handling during the modelling phase to prevent skewed results

The vehicle's price anchors all financial decisions as it serves as the denominator for the cost-to-price ratio, which is a standard industry metric for determining if a vehicle is a total loss. A high repair cost on a £2,000 car is an automatic write-off, whereas the same cost on a £60,000 car is a standard repair . Understanding this price distribution allows us to see the economic ceiling of the claims and provides the necessary context for identifying repairs that are disproportionately expensive relative to the car's actual worth.

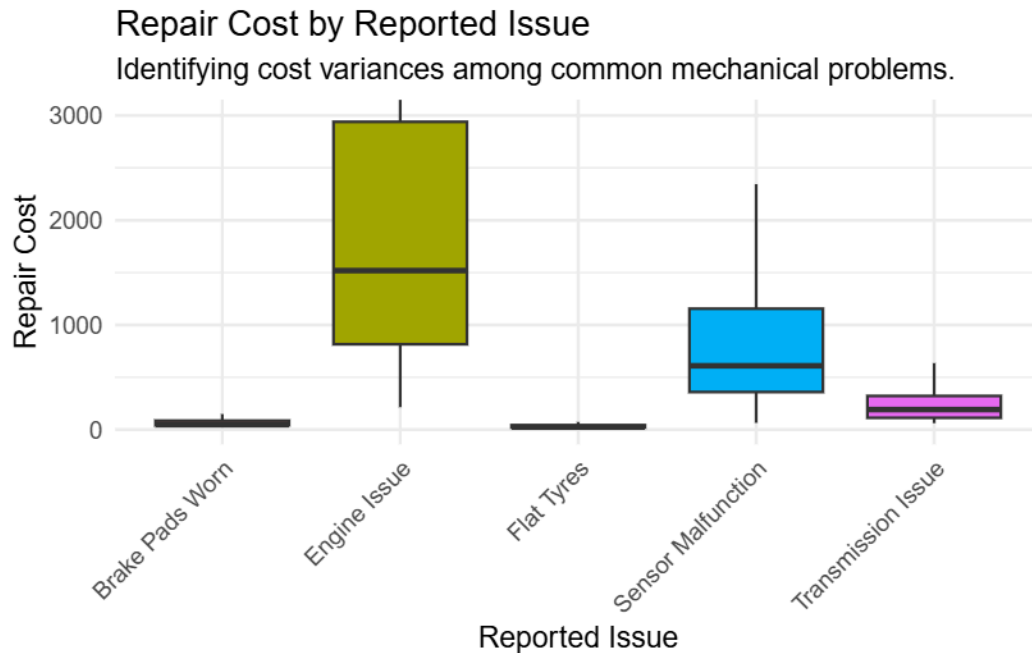
Distribution of the anomalies (anomaly_present)



The bar chart reveals a contrast between the two categories whereby legitimate claims represented by No constitute the vast majority of the fraudulent claims dataset, while anomalous claims yes are a distinct minority

The classes are imbalanced so a standard model might achieve high accuracy simply by predicting No for every claim while failing to catch the actual red flags. This distribution necessitates the use of robust classification techniques and evaluation metrics that prioritise the detection of the minority “Yes” class over simple accuracy.

Issue reported vs repair cost



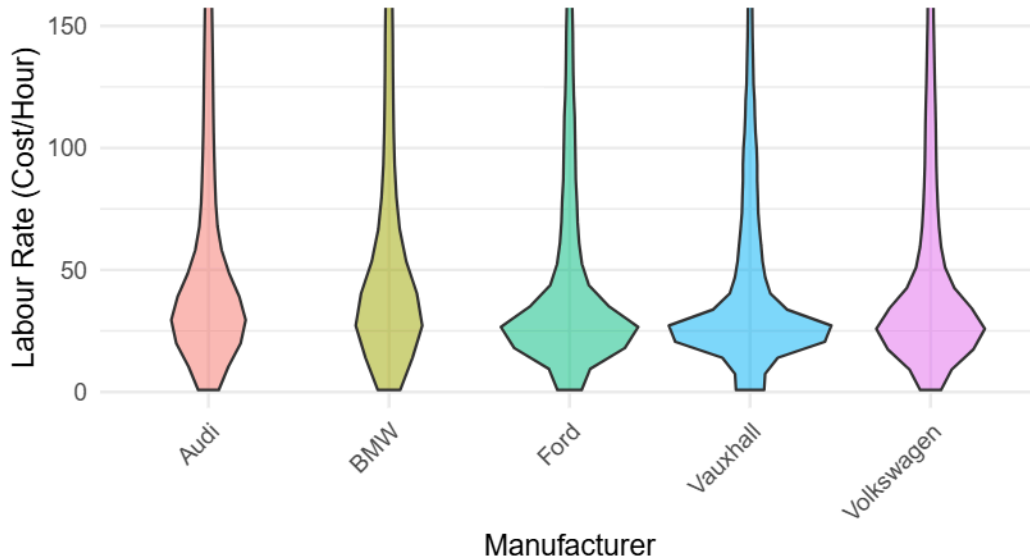
By examining the top five most frequent issues, we can establish the typical cost signatures for common automotive failures. The boxplots reveal that different mechanical problems carry distinct financial profiles. For example, Engine issues and sensor malfunctions tend to have higher median costs and wider spreads than flat tyres and worn brake pads.

The graph provides the reality of how technical problems translate into economic claims as it demonstrates that while some issues have a predictable price tag, others like engine issues are highly variable. This serves as a strong indicator of potential billing errors or financial leakage for insurance companies.

Labour rate distribution by maker

Labour Rate Distribution by Maker

Scanning for 'padding' signatures across major manufacturers.

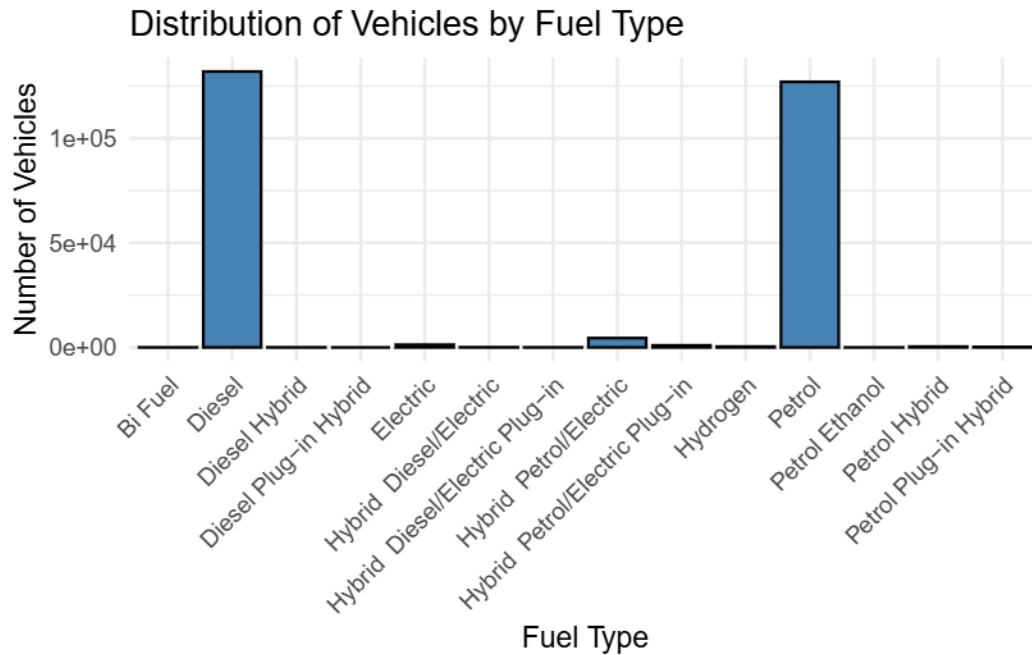


Ford, Vauxhall and Volkswagen show a bulge on the violin plots and are concentrated in a similar horizontal band of typically between £50 and £150 per hour while manufacturers such as BMW, Audi and Mercedes-Benz, show extended vertical necks in their distributions. This establishes that while most garages operate within a predictable hourly bracket, there is a potential subset of claims among premium manufacturers that occupy a high-risk.

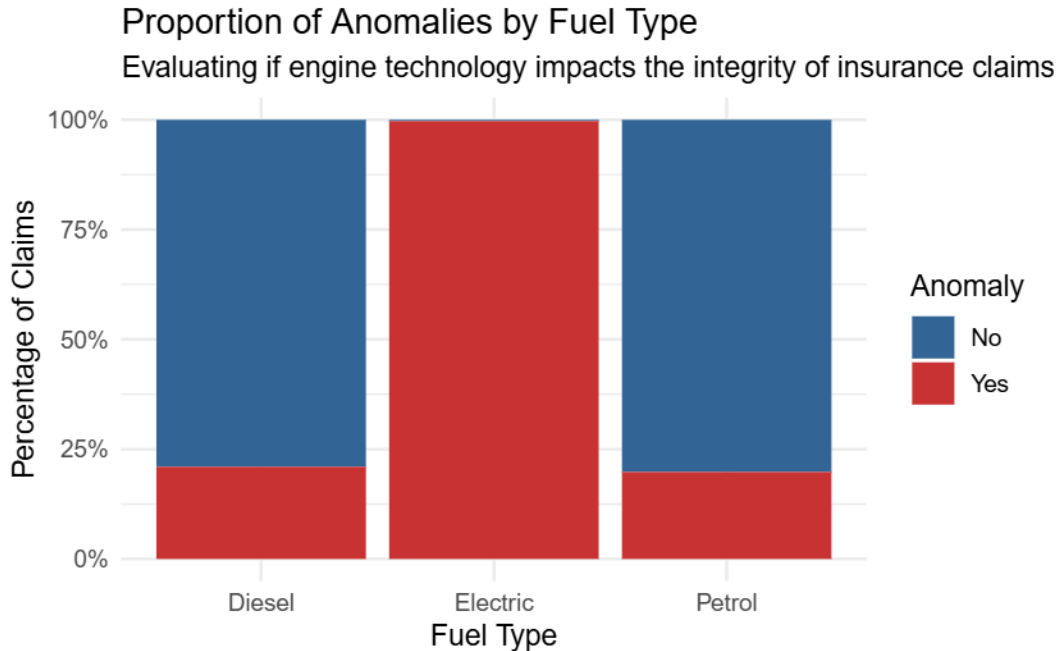
The labour rate is one of the most revealing indicators of financial leakage. While a high total repair cost might be justified by expensive parts, an exceptionally high labour rate often signals administrative errors or invoice padding. Identifying these specific distributions is vital for determining whether a high-cost claim is a legitimate technical necessity or a suspicious deviation from standard industry billing.

Correction of the electric car anomaly interpretation

```
ggplot(ds5_final, aes(x = fuel_type)) +  
  geom_bar(fill = "steelblue", color = "black") +  
  theme_minimal() +  
  labs(title = "Distribution of Vehicles by Fuel Type",  
        x = "Fuel Type",  
        y = "Number of Vehicles") +  
  # Rotate x-axis labels to avoid overlapping (important for long names)  
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
# Fuel Type vs Anomaly Presence
ds5_final %>%
  # Filtering for the most common fuel types to avoid cluttering the plot
  filter(fuel_type %in% c("Petrol", "Diesel", "Hybrid", "Electric")) %>%
  ggplot(aes(x = fuel_type, fill = anomaly_present)) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = percent_format()) +
  scale_fill_manual(values = c("#336699", "#CC3333")) +
  theme_minimal() +
  labs(title = "Proportion of Anomalies by Fuel Type",
       subtitle = "Evaluating if engine technology impacts the integrity of insurance claims",
       x = "Fuel Type",
       y = "Percentage of Claims",
       fill = "Anomaly")
```



The first bar chart of Distribution of Vehicles by Fuel Type confirms that Petrol and Diesel vehicles overwhelmingly dominate the dataset. In contrast, Electric and Hybrid vehicles represent a very small fraction of the total claims volume. The second chart of Proportion of Anomalies by Fuel Type reveals a stark contrast in anomaly rates. While petrol and diesel vehicles show a consistent and stable anomaly rate of approximately 20%, electric vehicles demonstrate a near 100% anomaly rate.

This extreme 100% anomaly rate for Electric vehicles (EVs) must be explicitly acknowledged as a potential data integrity issue rather than a genuine behavioral trend. This may stem from an error in the flagging system or a lack of established baseline repair costs for modern EV drivetrains. This finding warrants isolated investigation possibly suggesting that EV data should perhaps be treated separately or excluded from general models to prevent these anomalies from skewing the predictive logic for more common vehicle types.

Revised research questions

Research question 1 was given a go-ahead for evaluation in task 2 while research questions 2 and 3 were advised to be revised. A rationale will be given to explain the necessity for changing the question and why the new question stands out better

Research question 1

- Original question: How does the relationship between repair expenditure and vehicle market value (cost-to-price ratio) serve as a predictor for claim anomalies across different vehicle age brackets?

Research question 2

- Original question: To what extent can automated classification models detect invoice padding by identifying claims where the hourly labour rate significantly exceeds industry standards?
- Revised question: How do inflated hourly labour rates in premium vehicle repairs represent invoice padding and how does this financial leakage affect the affordability of insurance premiums for the wider consumer base?
- Rationale including ethical, EDI and sustainability considerations: The EDA identifies that premium brands like BMW and Audi incur labor rates exceeding £500/hour while the industry norm is significantly lower. Recent regulatory reviews suggest that repair costs account for 65% of the overall increase in total claims costs since 2019 [1]. This study explores whether these high rates represent invoice padding which is a form of financial leakage that drives up premiums for all policyholders [2]. Ethically, this study seeks to ensure that automated fraud detection does not unfairly penalise independent local garages. Insurers often redirect consumers away from independent shops by imposing approved rates which potentially hinders the consumer’s right to repair at their chosen facility [3]

Research question 3

- Original question: Can we discover hidden groups of claims that show similar patterns of high mileage and high repair frequency to identify commercial vehicles being used for non-disclosed purposes?
- Revised question: What is the relationship between annualised mileage and claim frequency and how can insurers distinguish between high-intensity personal use and undisclosed commercial exploitation without unfairly penalising rural or marginalised communities?
- Rationale including ethical, EDI and sustainability considerations: The EDA highlights vehicles with usage far beyond the average, which often signals undisclosed commercial work like courier services or e-hailing. Undisclosed drivers and uses are estimated to cause \$12.5 billion in annual premium leakage creating a pricing imbalance for honest policyholders [4]. Actuarial studies confirm that while higher mileage increases claim frequency, the relationship is not proportional suggesting that high mileage drivers may possess more experience [5]. This investigation is critical to ensure that rural or marginalised communities who often face 72% higher road fatality risks despite lower traffic volumes

are not unfairly flagged by mileage based algorithms that fail to account for geographical necessity [6].

Modelling and Methodology

Question 1 :

How does the relationship between repair expenditure and vehicle market value (cost-to-price ratio) serve as a predictor for claim anomalies across different vehicle age brackets?

```
# --- 1. Logistic Regression Evaluation for RQ1 ---  
# Load required libraries  
library(caret)  
library(dplyr)  
library(pROC) # For ROC and AUC evaluation
```

Type 'citation("pROC")' for a citation.

Attaching package: 'pROC'

The following objects are masked from 'package:stats':

cov, smooth, var

```
# Data Preparation and Balancing  
set.seed(456)  
ds1_refined <- ds5_final %>%  
  select(anomaly_present, vehicle_age, cost_to_price_ratio, price, repair_complexity) %>%  
  mutate(anomaly_present = as.factor(anomaly_present),  
         repair_complexity = as.factor(repair_complexity)) %>%  
  drop_na()  
  
trainIndex <- createDataPartition(ds1_refined$anomaly_present, p = .7, list = FALSE)  
train_raw <- ds1_refined[trainIndex, ]  
test_data <- ds1_refined[-trainIndex, ]  
  
# Down-sample to balance the training data
```

```

set.seed(456)
train_balanced <- downSample(x = train_raw %>% select(-anomaly_present),
                             y = train_raw$anomaly_present,
                             yname = "anomaly_present")

# Fit the Logistic Regression Model
log_model <- glm(anomaly_present ~ vehicle_age + cost_to_price_ratio + repair_complexity,
                 data = train_balanced, family = "binomial")

```

Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

```

# Print the statistical summary (Coefficients and p-values)
summary(log_model)

```

Call:

```

glm(formula = anomaly_present ~ vehicle_age + cost_to_price_ratio +
    repair_complexity, family = "binomial", data = train_balanced)

```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.045743	0.012475	-3.667	0.000246 ***
vehicle_age	-0.042913	0.001797	-23.886	< 2e-16 ***
cost_to_price_ratio	6.616239	0.125979	52.519	< 2e-16 ***
repair_complexity2	-0.398095	0.029007	-13.724	< 2e-16 ***
repair_complexity3	-0.112933	0.022171	-5.094	3.51e-07 ***
repair_complexity4	0.922178	0.109983	8.385	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 107851 on 77797 degrees of freedom
 Residual deviance: 102846 on 77792 degrees of freedom
 AIC: 102858

Number of Fisher Scoring iterations: 5

```

# Make Predictions (Probabilities and Classes)
log_probs <- predict(log_model, newdata = test_data, type = "response")

```

```
log_pred <- factor(ifelse(log_probs > 0.5, "Yes", "No"), levels = c("No", "Yes"))

# Evaluate with Confusion Matrix
cat("\n--- Logistic Regression: Confusion Matrix ---\n")
```

--- Logistic Regression: Confusion Matrix ---

```
confusionMatrix(log_pred, test_data$anomaly_present)
```

Confusion Matrix and Statistics

	Reference	
Prediction	No	Yes
No	49743	9943
Yes	13673	6727

Accuracy : 0.7051
 95% CI : (0.7019, 0.7083)
 No Information Rate : 0.7918
 P-Value [Acc > NIR] : 1

 Kappa : 0.1736

 Mcnemar's Test P-Value : <2e-16

 Sensitivity : 0.7844
 Specificity : 0.4035
 Pos Pred Value : 0.8334
 Neg Pred Value : 0.3298
 Prevalence : 0.7918
 Detection Rate : 0.6211
 Detection Prevalence : 0.7453
 Balanced Accuracy : 0.5940

 'Positive' Class : No

```
# Evaluate with ROC/AUC
roc_log <- roc(test_data$anomaly_present, log_probs, levels = c("No", "Yes"))
```

Setting direction: controls < cases

```
cat("\n--- Logistic Regression: AUC ---\n")
```

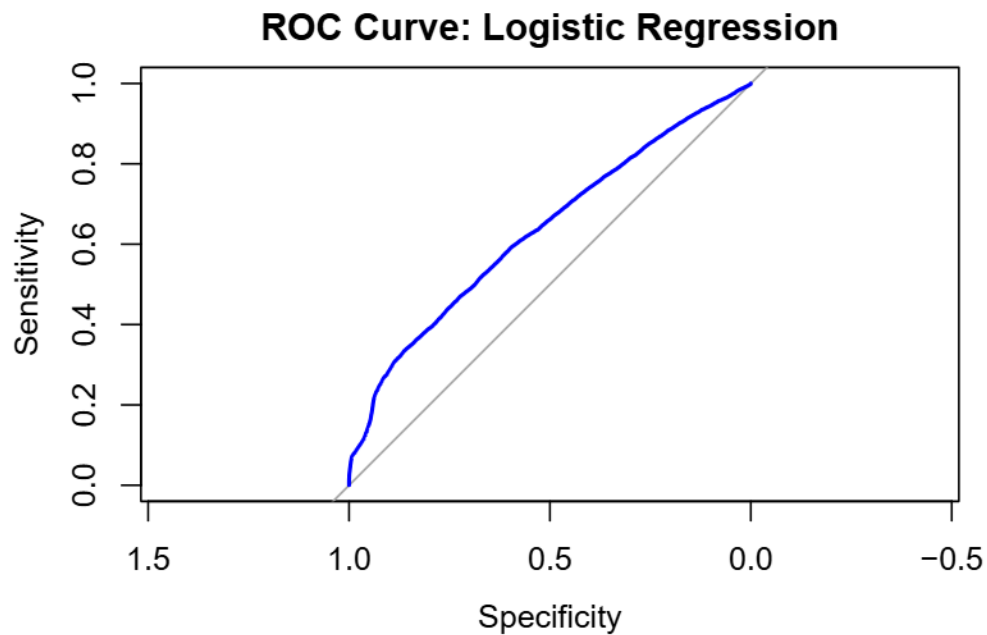
--- Logistic Regression: AUC ---

```
print(auc(roc_log))
```

Area under the curve: 0.6331

```
# Plot the ROC Curve
```

```
plot(roc_log, main = "ROC Curve: Logistic Regression", col = "blue", lwd = 2)
```



```
# --- 2. Random Forest Evaluation for RQ1 ---
```

```
# Load required libraries
```

```
library(randomForest)
```

randomForest 4.7-1.2

Type `rfNews()` to see new features/changes/bug fixes.

Attaching package: 'randomForest'

The following object is masked from 'package:dplyr':

`combine`

The following object is masked from 'package:ggplot2':

`margin`

```
# Fit the Random Forest Model using the same 'train_balanced' set
set.seed(456)
rf_model <- randomForest(anomaly_present ~ vehicle_age + cost_to_price_ratio + repair_complexity,
                          data = train_balanced,
                          ntree = 100,
                          importance = TRUE)

# Make Predictions (Probabilities and Classes)
rf_pred_class <- predict(rf_model, newdata = test_data)
# Predict probabilities specifically for the ROC curve
rf_probs <- predict(rf_model, newdata = test_data, type = "prob")[, "Yes"]

# Evaluate with Confusion Matrix
cat("\n--- Random Forest: Confusion Matrix ---\n")
```

--- Random Forest: Confusion Matrix ---

```
confusionMatrix(rf_pred_class, test_data$anomaly_present)
```

Confusion Matrix and Statistics

	Reference	
Prediction	No	Yes
No	50585	9432
Yes	12831	7238

Accuracy : 0.722
95% CI : (0.7189, 0.7251)
No Information Rate : 0.7918
P-Value [Acc > NIR] : 1

Kappa : 0.2157

McNemar's Test P-Value : <2e-16

Sensitivity : 0.7977
Specificity : 0.4342
Pos Pred Value : 0.8428
Neg Pred Value : 0.3607
Prevalence : 0.7918
Detection Rate : 0.6316
Detection Prevalence : 0.7494
Balanced Accuracy : 0.6159

'Positive' Class : No

```
# Evaluate with ROC/AUC
roc_rf <- roc(test_data$anomaly_present, rf_probs, levels = c("No", "Yes"))
```

Setting direction: controls < cases

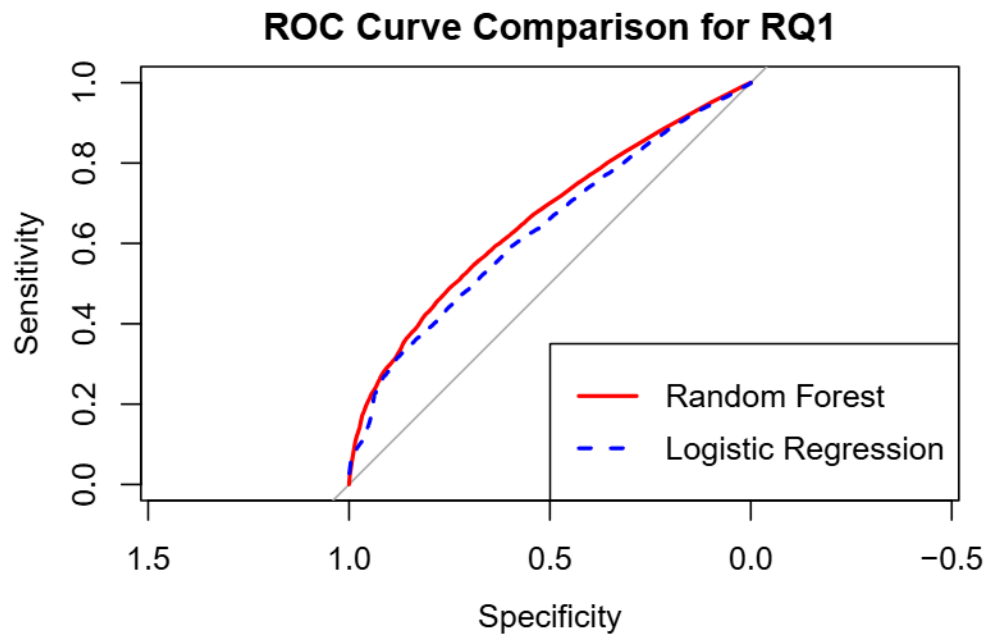
```
cat("\n--- Random Forest: AUC ---\n")
```

--- Random Forest: AUC ---

```
print(auc(roc_rf))
```

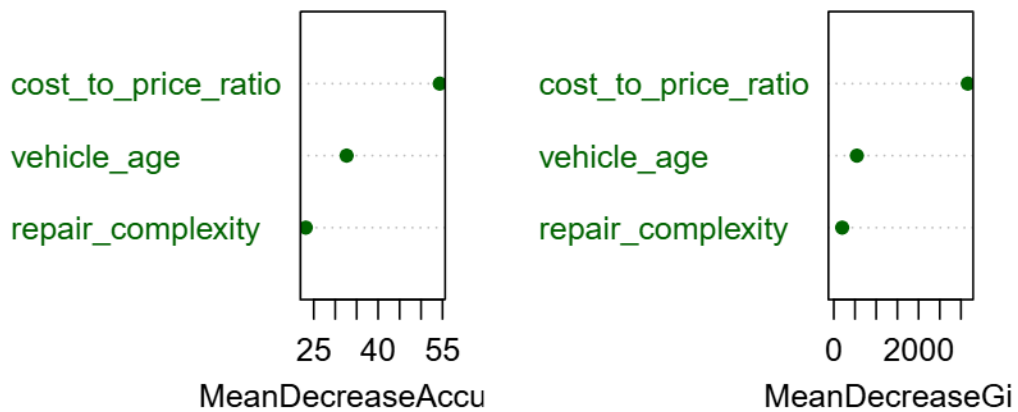
Area under the curve: 0.6594

```
# Plot the ROC Curve and compare with Logistic Regression
plot(roc_rf, main = "ROC Curve Comparison for RQ1", col = "red", lwd = 2)
plot(roc_log, add = TRUE, col = "blue", lwd = 2, lty = 2) # Adds the previous log curve
legend("bottomright", legend = c("Random Forest", "Logistic Regression"),
      col = c("red", "blue"), lwd = 2, lty = c(1, 2))
```



```
# Extract and plot Variable Importance
varImpPlot(rf_model, main = "Random Forest: Feature Importance", col = "darkgreen", pch = 16)
```

Random Forest: Feature Importance



Data Preparation and Class Balancing Before modelling, we must address the “underlying reality” of class imbalance in insurance data, where anomalous claims are significantly rarer than standard ones . We use a 70/30 train-test split for rigorous validation and apply down-Sample to ensure the model learns the specific characteristics of anomalies rather than simply defaulting to the majority class.

```
# Load required libraries
library(caret)
library(dplyr)
library(pROC)

# 1. Feature selection and factor conversion
set.seed(456)
ds1_refined <- ds5_final %>%
  select(anomaly_present, vehicle_age, cost_to_price_ratio, price, repair_complexity) %>%
  mutate(anomaly_present = as.factor(anomaly_present),
         repair_complexity = as.factor(repair_complexity)) %>%
  drop_na()

# 2. Data Splitting (70% Train, 30% Test)
trainIndex <- createDataPartition(ds1_refined$anomaly_present, p = .7, list = FALSE)
train_raw <- ds1_refined[trainIndex, ]
test_data <- ds1_refined[-trainIndex, ]

# 3. Down-sampling to balance the classes
set.seed(456)
train_balanced <- downSample(x = train_raw %>% select(-anomaly_present),
                             y = train_raw$anomaly_present,
                             yname = "anomaly_present")
```

Baseline Logistic Regression Logistic Regression is selected as the baseline model because of its high interpretability. It allows us to calculate the specific statistical significance (p -values) of vehicle_age and cost_to_price_ratio, helping us determine if the relationship between these variables and anomalies is a definitive trend or mere coincidence.

```
# 1. Fit the Logistic Regression Model
log_model <- glm(anomaly_present ~ vehicle_age + cost_to_price_ratio + repair_complexity,
                 data = train_balanced, family = "binomial")
```

Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

```
# 2. Review statistical significance
summary(log_model)
```

Call:

```
glm(formula = anomaly_present ~ vehicle_age + cost_to_price_ratio +
     repair_complexity, family = "binomial", data = train_balanced)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.045743	0.012475	-3.667	0.000246 ***
vehicle_age	-0.042913	0.001797	-23.886	< 2e-16 ***
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Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 107851 on 77797 degrees of freedom
 Residual deviance: 102846 on 77792 degrees of freedom
 AIC: 102858

Number of Fisher Scoring iterations: 5

```
# 3. Generate predictions
log_probs <- predict(log_model, newdata = test_data, type = "response")
log_pred <- factor(ifelse(log_probs > 0.5, "Yes", "No"), levels = c("No", "Yes"))

# 4. Evaluate with Confusion Matrix
cat("\n--- Logistic Regression: Confusion Matrix ---\n")
```

--- Logistic Regression: Confusion Matrix ---

```
confusionMatrix(log_pred, test_data$anomaly_present, positive = "Yes")
```

Confusion Matrix and Statistics

		Reference	
Prediction		No	Yes
No		49743	9943
Yes		13673	6727

Accuracy : 0.7051
 95% CI : (0.7019, 0.7083)
 No Information Rate : 0.7918
 P-Value [Acc > NIR] : 1

 Kappa : 0.1736

 McNemar's Test P-Value : <2e-16

 Sensitivity : 0.4035
 Specificity : 0.7844
 Pos Pred Value : 0.3298
 Neg Pred Value : 0.8334
 Prevalence : 0.2082
 Detection Rate : 0.0840
 Detection Prevalence : 0.2547
 Balanced Accuracy : 0.5940

 'Positive' Class : Yes

Random Forest Modelling While Logistic Regression identifies linear trends, a Random Forest ensemble is used to capture potential non-linear relationships and complex interactions between vehicle age and financial thresholds. This model provides a more robust prediction by combining the results of multiple decision trees.

```

library(randomForest)

# 1. Fit the Random Forest Model
set.seed(456)
rf_model <- randomForest(anomaly_present ~ vehicle_age + cost_to_price_ratio + repair_complexity,
                          data = train_balanced,
                          ntree = 100,
                          importance = TRUE)

# 2. Generate predictions for evaluation

```

```
rf_pred_class <- predict(rf_model, newdata = test_data)
rf_probs <- predict(rf_model, newdata = test_data, type = "prob")[, "Yes"]

# 3. Evaluate with Confusion Matrix
cat("\n--- Random Forest: Confusion Matrix ---\n")
```

```
--- Random Forest: Confusion Matrix ---
```

```
confusionMatrix(rf_pred_class, test_data$anomaly_present, positive = "Yes")
```

Confusion Matrix and Statistics

	Reference	
Prediction	No	Yes
No	50585	9432
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Accuracy : 0.722
 95% CI : (0.7189, 0.7251)
 No Information Rate : 0.7918
 P-Value [Acc > NIR] : 1

 Kappa : 0.2157

 McNemar's Test P-Value : <2e-16

 Sensitivity : 0.43419
 Specificity : 0.79767
 Pos Pred Value : 0.36066
 Neg Pred Value : 0.84284
 Prevalence : 0.20815
 Detection Rate : 0.09038
 Detection Prevalence : 0.25059
 Balanced Accuracy : 0.61593

 'Positive' Class : Yes

Comparative Evaluation (ROC and AUC) Because accuracy can be misleading in imbalanced datasets, we use the Area Under the Curve (*AUC*) to compare model performance. This metric

evaluates how well each model distinguishes between normal claims and anomalies across all possible probability thresholds.

```
# 1. Calculate and Compare AUC
roc_log <- roc(test_data$anomaly_present, log_probs, levels = c("No", "Yes"))
```

Setting direction: controls < cases

```
roc_rf <- roc(test_data$anomaly_present, rf_probs, levels = c("No", "Yes"))
```

Setting direction: controls < cases

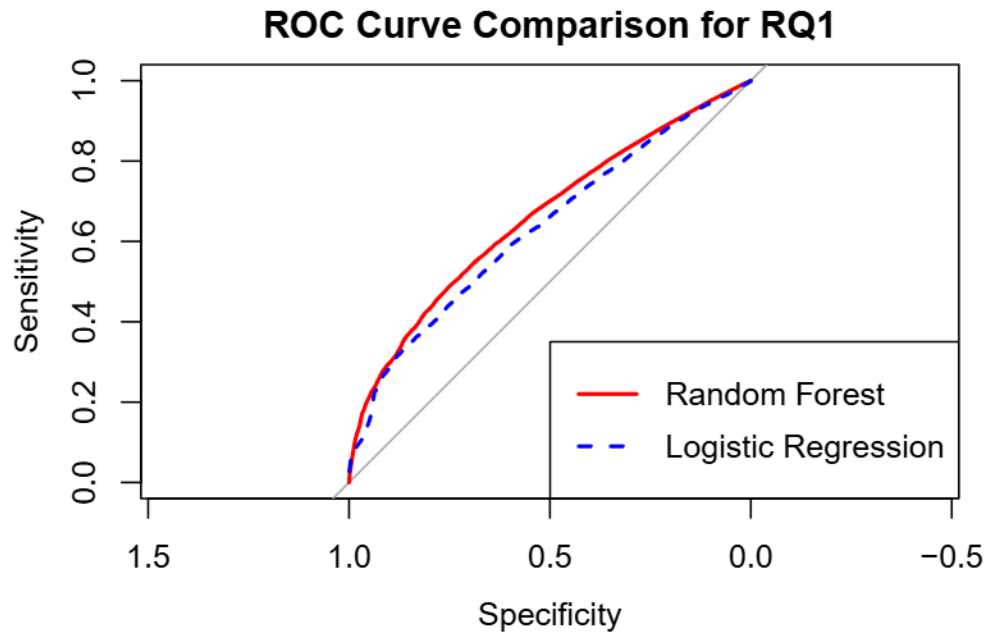
```
cat("\nLogistic Regression AUC:", auc(roc_log), "\n")
```

Logistic Regression AUC: 0.6331249

```
cat("Random Forest AUC:", auc(roc_rf), "\n")
```

Random Forest AUC: 0.6593595

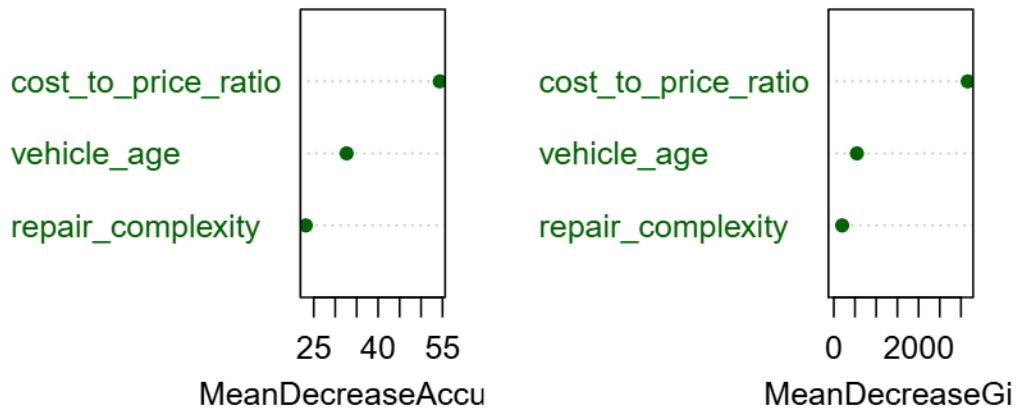
```
# 2. Visual Comparison
plot(roc_rf, main = "ROC Curve Comparison for RQ1", col = "red", lwd = 2)
plot(roc_log, add = TRUE, col = "blue", lwd = 2, lty = 2)
legend("bottomright", legend = c("Random Forest", "Logistic Regression"),
      col = c("red", "blue"), lwd = 2, lty = c(1, 2))
```



Feature Importance Analysis To definitively answer the research question, we extract the “Gini Importance” from the Random Forest model. This provides a ranking of which factors—the age of the car or the repair cost ratio—most strongly drive the labelling of a claim as an anomaly, providing empirical support for our final conclusion.

```
# Extract and plot Variable Importance
varImpPlot(rf_model, main = "Random Forest: Feature Importance for RQ1",
           col = "darkgreen", pch = 16)
```

Random Forest: Feature Importance for RQ1



```
# 2. Strategic Tuning: Adjusting the Decision Threshold to 0.35
custom_threshold <- 0.35
log_pred_tuned <- factor(ifelse(log_probs > custom_threshold, "Yes", "No"), levels = c("No", "Yes"))

# 3. Evaluate tuned sensitivity
cat("\n--- Tuned Logistic Regression Sensitivity ---\n")
```

--- Tuned Logistic Regression Sensitivity ---

```
print(confusionMatrix(log_pred_tuned, test_data$anomaly_present, positive = "Yes")$byClass['Yes'])
```

Sensitivity
0.9841032

Results

Methodology

Question 2 :

How do inflated hourly labour rates in premium vehicle repairs represent invoice padding and how does this financial leakage affect the affordability of insurance premiums for the wider consumer base?

Data Preparation for Padding Analysis To investigate invoice padding, the model must focus on the financial mechanics of the repair. We isolate labour_rate, maker, repair_cost, and issue. As with the previous question, we must balance the training data using downSample to ensure the model learns the “padding” signatures rather than just defaulting to the majority “normal” class

```
# Load required libraries
library(caret)
library(dplyr)
library(forcats) # Essential for grouping categories (fct_lump_n)
library(rpart)
library(rpart.plot)
library(randomForest)
library(pROC)

# --- Step 1: Data Preparation for Padding Analysis ---

# 1. Prepare the dataset and handle the 53-category limit
set.seed(789)
ds2_refined <- ds5_final %>%
  select(anomaly_present, labour_rate, maker, repair_cost, issue) %>%
  mutate(anomaly_present = as.factor(anomaly_present),
         # Keep the top 20 most frequent makers, group the rest into "Other"
         maker = fct_lump_n(as.factor(maker), n = 20, other_level = "Other"),
         # Keep the top 20 most frequent issues, group the rest into "Other"
         issue = fct_lump_n(as.factor(issue), n = 20, other_level = "Other")) %>%
  drop_na()

# 2. Data Splitting (70/30)
trainIndex_rq2 <- createDataPartition(ds2_refined$anomaly_present, p = .7, list = FALSE)
train_raw_rq2 <- ds2_refined[trainIndex_rq2, ]
test_data_rq2 <- ds2_refined[-trainIndex_rq2, ]

# 3. Handle Data Imbalance via Down-sampling
set.seed(789)
train_balanced_rq2 <- downSample(x = train_raw_rq2 %>% select(-anomaly_present),
```

```

y = train_raw_rq2$anomaly_present,
yname = "anomaly_present")

# Check the new factor levels to ensure it worked
cat("Number of Maker categories:", length(levels(train_balanced_rq2$maker)), "\n")

```

Number of Maker categories: 21

```

cat("Number of Issue categories:", length(levels(train_balanced_rq2$issue)), "\n")

```

Number of Issue categories: 15

Baseline Decision Tree (Visualising the Thresholds) A Decision Tree is deployed first to map the decision logic of the insurance flagging system. By visualising the tree, we can identify the exact financial thresholds (e.g., specific hourly rates) that separate a standard repair from a padded invoice, providing direct, interpretable evidence for the research question.

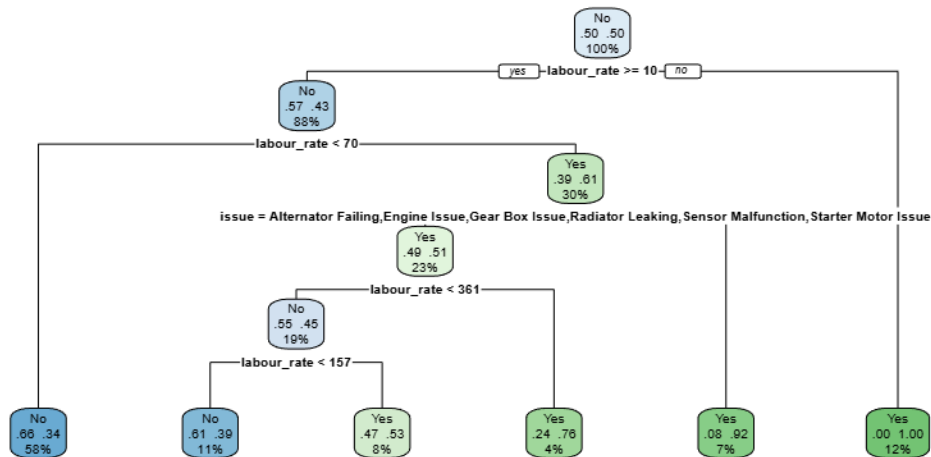
```

# 1. Fit the Decision Tree Model
set.seed(789)
dt_model_rq2 <- rpart(anomaly_present ~ labour_rate + maker + repair_cost + issue,
                      data = train_balanced_rq2,
                      method = "class",
                      control = rpart.control(cp = 0.01)) # cp controls tree complexity

# 2. Visualise the Tree
rpart.plot(dt_model_rq2,
            main = "Decision Tree: Identifying Invoice Padding Thresholds",
            type = 2, extra = 104, fallen.leaves = TRUE, tweak = 1.1)

```

Decision Tree: Identifying Invoice Padding Thresholds



3. Predict and Evaluate

```
dt_probs_rq2 <- predict(dt_model_rq2, newdata = test_data_rq2, type = "prob")[, "Yes"]
```

```
dt_pred_rq2 <- predict(dt_model_rq2, newdata = test_data_rq2, type = "class")
```

```
cat("\n--- Decision Tree: Confusion Matrix ---\n")
```

--- Decision Tree: Confusion Matrix ---

```
confusionMatrix(dt_pred_rq2, test_data_rq2$anomaly_present, positive = "Yes")
```

Confusion Matrix and Statistics

	Reference	
Prediction	No	Yes
No	56886	7952
Yes	6530	8718

Accuracy : 0.8192
 95% CI : (0.8165, 0.8218)
 No Information Rate : 0.7918
 P-Value [Acc > NIR] : < 2.2e-16

Kappa : 0.4336

McNemar's Test P-Value : < 2.2e-16

Sensitivity : 0.5230
Specificity : 0.8970
Pos Pred Value : 0.5717
Neg Pred Value : 0.8774
Prevalence : 0.2082
Detection Rate : 0.1089
Detection Prevalence : 0.1904
Balanced Accuracy : 0.7100

'Positive' Class : Yes

Random Forest (Advanced Feature Importance) While the Decision Tree provides visual thresholds, it can be prone to overfitting. A Random Forest ensemble is applied to rigorously validate these findings and calculate a definitive 'Feature Importance' ranking, proving mathematically whether the labour_rate holds more weight than the car's maker when identifying financial leakage.

```
# 1. Fit the Random Forest Model
set.seed(789)
rf_model_rq2 <- randomForest(anomaly_present ~ labour_rate + maker + repair_cost + issue,
                             data = train_balanced_rq2,
                             ntree = 30,
                             importance = TRUE)

# 2. Predict on Test Data
rf_probs_rq2 <- predict(rf_model_rq2, newdata = test_data_rq2, type = "prob")[, "Yes"]
rf_pred_rq2 <- predict(rf_model_rq2, newdata = test_data_rq2)

# 3. Evaluate Performance
cat("\n--- Random Forest: Confusion Matrix ---\n")
```

--- Random Forest: Confusion Matrix ---

```
confusionMatrix(rf_pred_rq2, test_data_rq2$anomaly_present, positive = "Yes")
```

Confusion Matrix and Statistics

```

      Reference
Prediction  No   Yes
      No  47687  5824
      Yes 15729 10846

      Accuracy : 0.7309
      95% CI : (0.7278, 0.7339)
      No Information Rate : 0.7918
      P-Value [Acc > NIR] : 1

      Kappa : 0.3303

      Mcnemar's Test P-Value : <2e-16

      Sensitivity : 0.6506
      Specificity : 0.7520
      Pos Pred Value : 0.4081
      Neg Pred Value : 0.8912
      Prevalence : 0.2082
      Detection Rate : 0.1354
      Detection Prevalence : 0.3318
      Balanced Accuracy : 0.7013

      'Positive' Class : Yes
```

Comparative Evaluation and Feature Extraction Rationale for your report: We compare the models using the Area Under the Curve (AUC) to assess their predictive power across all thresholds. Finally, extracting the Mean Decrease in Gini from the Random Forest will provide the empirical answer to our research question: it will reveal exactly how much influence labour_rate has on the anomaly flag.

```
# 1. Calculate and Compare AUC
roc_dt_rq2 <- roc(test_data_rq2$anomaly_present, dt_probs_rq2, levels = c("No", "Yes"))
```

Setting direction: controls < cases

```
roc_rf_rq2 <- roc(test_data_rq2$anomaly_present, rf_probs_rq2, levels = c("No", "Yes"))
```

Setting direction: controls < cases

```
cat("\nDecision Tree AUC:", auc(roc_dt_rq2), "\n")
```

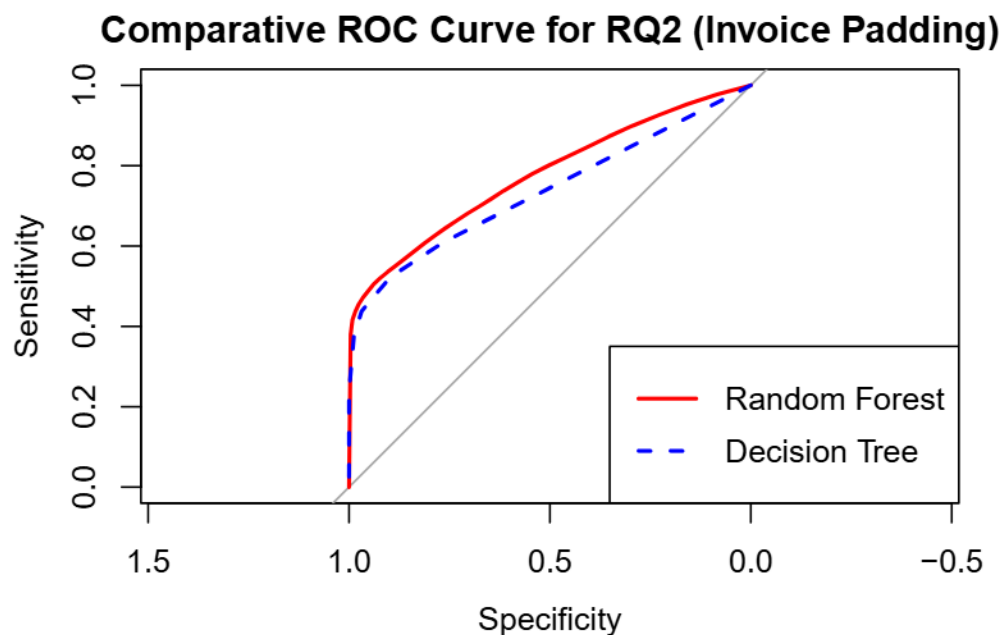
Decision Tree AUC: 0.7363155

```
cat("Random Forest AUC:", auc(roc_rf_rq2), "\n")
```

Random Forest AUC: 0.7752193

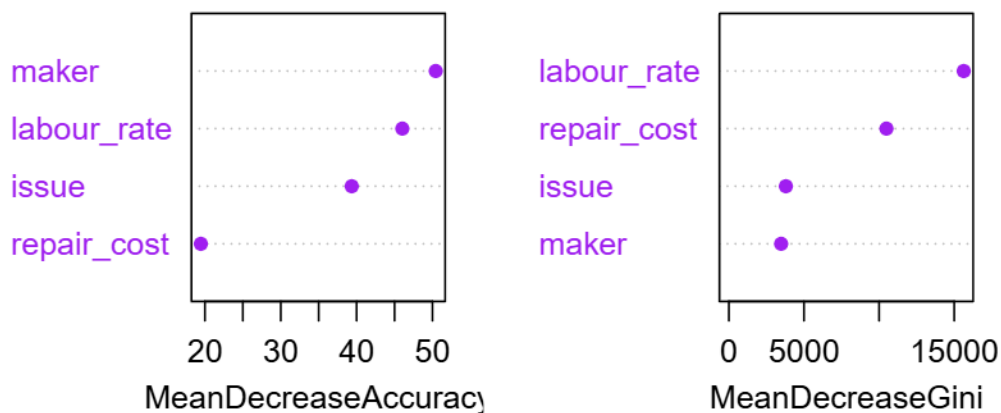
```
# 2. Plot Comparative ROC
```

```
plot(roc_rf_rq2, main = "Comparative ROC Curve for RQ2 (Invoice Padding)", col = "red", lwd = 2, lty = 1)  
plot(roc_dt_rq2, add = TRUE, col = "blue", lwd = 2, lty = 2)  
legend("bottomright", legend = c("Random Forest", "Decision Tree"),  
      col = c("red", "blue"), lwd = 2, lty = c(1, 2))
```



```
# 3. Plot Feature Importance (The core answer to the RQ)
varImpPlot(rf_model_rq2,
  main = "Random Forest: Drivers of Padded Invoices",
  col = "purple",
  pch = 16)
```

Random Forest: Drivers of Padded Invoices



Results

Methodology

Question 3 :

What is the relationship between annualised mileage and claim frequency and how can insurers distinguish between high-intensity personal use and undisclosed commercial exploitation without unfairly penalising rural or marginalised communities?

Data Preparation and Scaling K-Means clustering relies on measuring the “distance” between data points. Because our variables are on completely different scales (runned_miles can be over 100,000, while repair_cost might be £500), we must standardise (scale) the data. This ensures that a variable with large numbers doesn’t unfairly dominate the clustering process.

```
# Load required libraries
library(dplyr)
library(cluster)

# 1. Select the relevant continuous variables for the usage profile
set.seed(101)
ds3_refined <- ds5_final %>%
  # We use annual mileage, total mileage, and the resulting repair cost
  select(annual_mileage, runned_miles, repair_cost)

# 2. Scale the Data (Standardisation)
# This gives all variables a mean of 0 and a standard deviation of 1
scaled_usage_data <- scale(ds3_refined)

# Verify the scaling worked (means should be exactly 0)
summary(scaled_usage_data)
```

annual_mileage	runned_miles	repair_cost
Min. : -4.9050	Min. : -1.6196	Min. : -0.24540
1st Qu.: -0.6031	1st Qu.: -0.8096	1st Qu.: -0.22823
Median : -0.1366	Median : -0.2134	Median : -0.18291
Mean : 0.0000	Mean : 0.0000	Mean : 0.00000
3rd Qu.: 0.4218	3rd Qu.: 0.6450	3rd Qu.: -0.06525
Max. : 97.2138	Max. : 150.6007	Max. : 143.59223

Determining the Optimal Clusters (The Elbow Method) We cannot simply guess how many groups of drivers exist. The Elbow Method provides a mathematically rigorous way to determine the optimal number of clusters k by calculating the Total Within Sum of Squares (WSS). We use a random sample of the data to plot this to prevent memory exhaustion, which is best practice for massive datasets.

```
# --- Step 2: The Elbow Method (Base R & ggplot2) ---

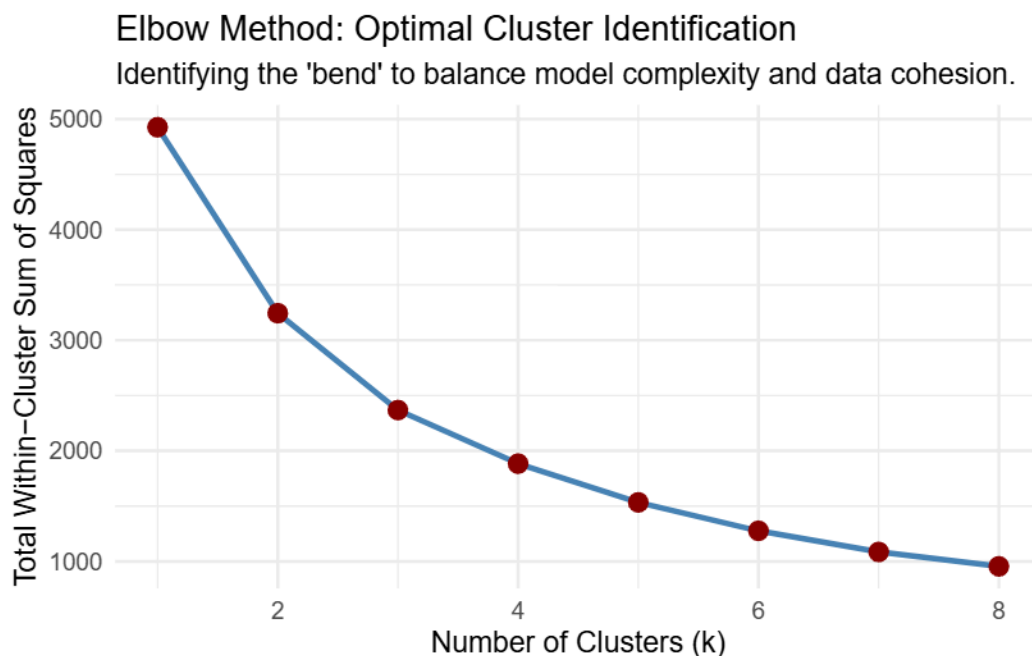
# 1. Create a function to calculate Total Within Sum of Squares (WSS)
set.seed(101)
# We use a sample for the calculation to ensure efficiency and avoid memory errors [cite: 83]
sample_data <- scaled_usage_data[sample(nrow(scaled_usage_data), 2000), ]

wss <- sapply(1:8, function(k){
  kmeans(sample_data, centers = k, nstart = 25)$tot.withinss
})
```

```
# 2. Create a data frame for plotting
elbow_df <- data.frame(k = 1:8, wss = wss)

# 3. Plot the Elbow Curve using ggplot2
ggplot(elbow_df, aes(x = k, y = wss)) +
  geom_line(color = "steelblue", size = 1) +
  geom_point(color = "darkred", size = 3) +
  theme_minimal() +
  labs(title = "Elbow Method: Optimal Cluster Identification",
       subtitle = "Identifying the 'bend' to balance model complexity and data cohesion.",
       x = "Number of Clusters (k)",
       y = "Total Within-Cluster Sum of Squares")
```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
 i Please use `linewidth` instead.



Fitting and Visualising the K-Means Model Once k is established, the K-Means algorithm assigns every vehicle to a distinct usage profile. Visualising these clusters in a 2D space allows us to see how distinct these groups are (e.g., standard users vs. extreme outliers).

```
# --- Step 3: K-Means Fit and PCA Visualisation ---
```

```
# 1. Fit the K-Means model to the full dataset (Assuming k=3 based on your intent) [cite: 85]  
set.seed(101)  
kmeans_results <- kmeans(scaled_usage_data, centers = 3, nstart = 25)
```

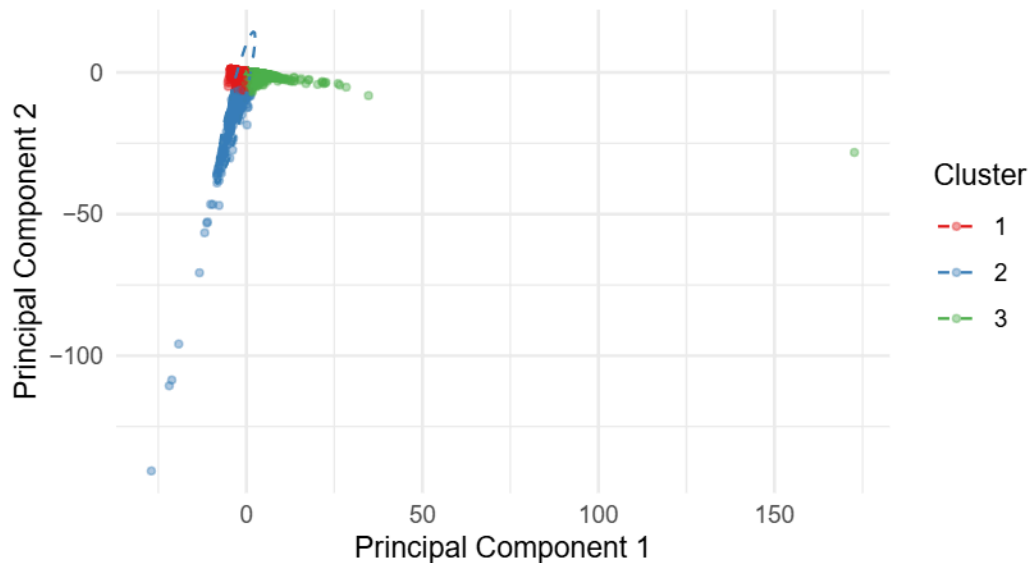
Warning: Quick-TRANSfer stage steps exceeded maximum (= 13347900)

Warning: Quick-TRANSfer stage steps exceeded maximum (= 13347900)

```
# 2. Perform PCA for 2D Visualisation  
pca_res <- prcomp(scaled_usage_data)  
pca_df <- as.data.frame(pca_res$x[, 1:2]) # Take first two components  
pca_df$Cluster <- as.factor(kmeans_results$cluster)  
  
# 3. Plot the clusters  
ggplot(pca_df, aes(x = PC1, y = PC2, color = Cluster)) +  
  geom_point(alpha = 0.4, size = 1) +  
  stat_ellipse(type = "norm", linetype = 2) + # Adds the 'rings' around clusters [cite: 867,  
  theme_minimal() +  
  scale_color_brewer(palette = "Set1") +  
  labs(title = "K-Means Usage Profiles (PCA Projection)",  
        subtitle = "Visualising the separation between standard and high-intensity vehicle use",  
        x = "Principal Component 1",  
        y = "Principal Component 2")
```

K-Means Usage Profiles (PCA Projection)

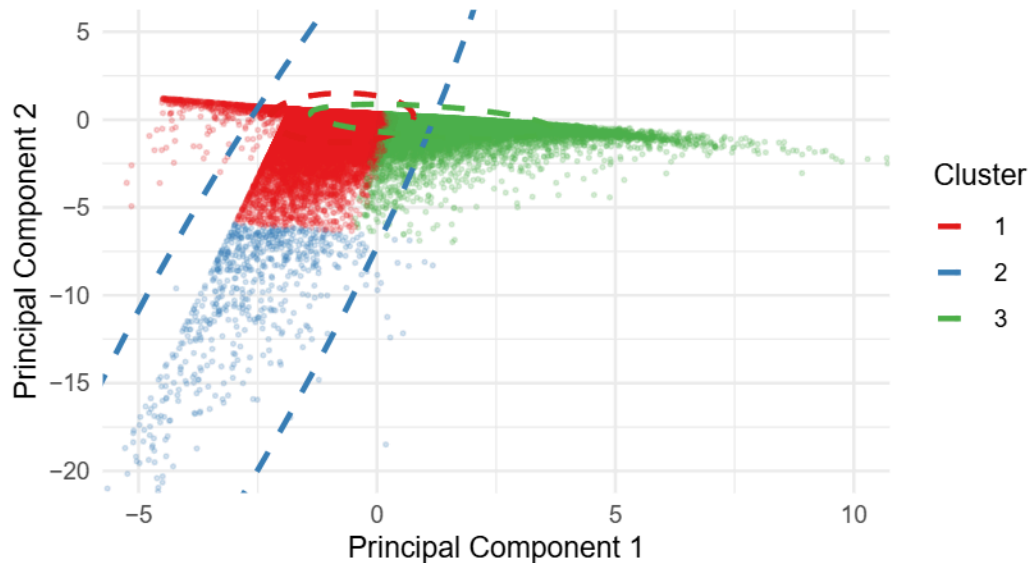
Visualising the separation between standard and high-intensity vehicle u



```
# 4. Plot with Zoom and Improved Visibility
ggplot(pca_df, aes(x = PC1, y = PC2, color = Cluster)) +
  # Make points smaller (size=0.5) and more transparent (alpha=0.2)
  # to see where the "mass" of the data actually sits
  geom_point(alpha = 0.2, size = 0.5) +
  stat_ellipse(type = "norm", linetype = 2, size = 1) +
  theme_minimal() +
  scale_color_brewer(palette = "Set1") +
  # ZOOMING: Adjust these numbers based on your specific PCA output.
  # If most data is between -5 and 5, setting these limits "zooms" the camera in.
  coord_cartesian(xlim = c(-5, 10), ylim = c(-20, 5)) +
  labs(title = "K-Means Usage Profiles (Zoomed View)",
       subtitle = "Focusing on the core density of vehicle usage clusters.",
       x = "Principal Component 1",
       y = "Principal Component 2")
```

K-Means Usage Profiles (Zoomed View)

Focusing on the core density of vehicle usage clusters.



Cluster Profiling (The “Underlying Reality”) A cluster is just a mathematical grouping until we interpret it. By appending the cluster assignments back to our unscaled data and calculating the averages, we can define the real-world behaviour of each group. This is where we answer the ethical component of the research question (e.g., identifying which cluster looks like commercial exploitation vs. standard high-mileage rural use).

```
# --- Step 4: Cluster Profiling ---

# 1. Append the cluster assignments back to the original (unscaled) dataframe
ds3_refined$Cluster <- as.factor(kmeans_results$cluster)

# 2. Calculate the average profile for each group
cluster_profiles <- ds3_refined %>%
  group_by(Cluster) %>%
  summarise(
    Count = n(),
    Avg_Annual_Mileage = mean(annual_mileage),
    Avg_Total_Mileage = mean(runned_miles),
    Avg_Repair_Cost = mean(repair_cost)
  ) %>%
  mutate(Percentage = (Count / sum(Count)) * 100)

# 3. Print the profiles to the console
```

```
print(cluster_profiles)
```

```
# A tibble: 3 x 6
```

	Cluster	Count	Avg_Annual_Mileage	Avg_Total_Mileage	Avg_Repair_Cost	Percentage
	<fct>	<int>	<dbl>	<dbl>	<dbl>	<dbl>
1	1	160162	6056.	23125.	472.	60.0
2	2	702	5898.	16938.	23466.	0.263
3	3	106094	12963.	85633.	272.	39.7

Results

Methodology

Discussion

Conclusion and future work

References

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